Aarush Mane

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EDUCATION

Princeton University — Princeton, NJ
Bachelor of Science in Engineering, Computer Science — Expected 05/2030
- Dual Enrollment: Single Variable Analysis (MAT215), Linear Algebra with Intro to Proofs (MAT217), 2025-2026.

Robbinsville High School — Graduated 06/2026
- Weighted GPA: 4.682/5.0; Unweighted: 3.977/4.0; SAT: 1530. National Merit Scholarship Winner; AP Scholar with Distinction; NJ State Seal of Biliteracy (Spanish).

SELECTED PROJECT

Nemo | Pollution Source Investigation — HackHarvard 2026
Simulation Workflow, Validation & Technical Pitching — 10/16-10/18/2026
- **3rd Overall Best Project; 2nd, ASUS "Save the Earth" Sustainability Track.** Pitched Nemo to multiple judges and sponsors, including **Capital One** and startup representatives.
- **Conceived the Grok-assisted physics workflow** and authored simulation instructions exploring **advection, diffusion, Runge-Kutta (RK4), and Kalman filtering** for waste transport and reuse on later samples.
- **Developed export-validation tests** to detect missing or contradictory trial records and verify **SHA-256 hashes**; co-designed reproducible protocols separating physical experiments from simulations.

EXPERIENCE

FIRST Robotics Competition Team 2590, Nemesis — Robbinsville, NJ
Software Development & Strategy Lead — 10/2023-06/2026
- Led technical strategy and statistical performance modeling across **65 members**; built Excel/Sheets analytics supporting a **5th-place finish among 600+ teams** at the World Championship.
- Built **Java/OpenCV vision**, Oculus Quest localization, and AprilTag PnP pose estimation; presented control systems to judges for the **Innovation in Control Award**.
- Developed Java trajectory wrappers and subsystem scheduling; built **Argus**, a Slack AI agent integrating MiniMax, Nanobot, Python API scrapers, and GitHub workflows for scouting analytics.
- Mentored **10+ developers** in Java, autonomous control, Git branching, testing, and pull request review.

NJ Governor's School of Engineering & Technology — Rutgers University, NJ
Research Scholar — 07/2025
- Collaborated on a **5-person research team** modeling autonomous path tracking; presented research to **300+ faculty, corporate sponsors, and industry partners** at the GSET Final Symposium.

FIRST Tech Challenge Team 22348, REAPR — Robbinsville, NJ
Founder & Software Mentor — 10/2022-06/2026
- Secured **$16,000+ in sponsorships** for a **15-member** community robotics initiative; managed institutional partnerships and expanded regional outreach and participation by **30%**.
- Taught TensorFlow object detection and Java autonomous routines; built AppSheet data pipelines for tactical decisions.

iStart Valley — Remote
Financial Projections & Business Strategy Intern — 01/2025-06/2025
- Built financial projections, unit economics, and TAM/SAM/SOM analysis for the **BioPack concept pitch** (modeled $500k valuation); earned **Global Semifinalist** placement in the startup pitch contest.
- Earned the **Gold Presidential Volunteer Service Award** for 265 hours of technical writing, stakeholder engagement, and community outreach.

PUBLICATIONS

IEEE MIT Undergraduate Research Technology Conference — Presented 10/2025
Co-Author: PID Control Optimization and Trajectory Tracking for Autonomous Differential Drive Robots | IEEE Xplore
- Co-developed a Python nonlinear PID controller with moving integral windows, velocity-dependent target yaws, and anti-windup limits; compared A*/Dijkstra paths and ROS GazeboSim LiDAR obstacle avoidance.
- Validated controller performance with **TurtleBot Vicon** tracking and matched-pairs t-tests (**$p < 0.01$**).

TECHNICAL SKILLS

Languages & Data: Java, Python, MATLAB, LaTeX, JSON, HTML/CSS, Excel, Google Sheets, hypothesis testing
Tools & Methods: OpenCV, TensorFlow, ROS, GazeboSim, Git, GitHub Actions, Vicon, AppSheet, SHA-256, software validation
Strategy: Sponsorship proposals, technical presentations, TAM/SAM/SOM | **Spoken:** English, Spanish, Telugu